CyberLink Networks

Securing Your Digital Infrastructure

Enterprise Network Infrastructure Project

Design and implementation of a secure, multi-branch enterprise network featuring dynamic routing, centralized services, internet connectivity simulation, and layered security controls — built in Cisco Packet Tracer.

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1 Introduction

This project presents the design and full implementation of a multi-branch enterprise network for **CyberLink Networks**, simulated entirely in Cisco Packet Tracer. The network connects five independent branches — Human Resources, Sales, Development, a central Data Center, and a Remote Office — through a central core router using dynamic routing. The project demonstrates practical, hands-on mastery of enterprise networking concepts including IP addressing (VLSM), dynamic routing, centralized network services, internet connectivity simulation, and layered network security.

Project Objectives

- **Connectivity:** Establish full routed connectivity between five branches through a central hub router.
- **Automation:** Deploy centralized DHCP for automatic IP assignment across all branches.
- **Name Resolution:** Deploy a DNS server to resolve internal and external domain names.
- **Hosting:** Host an internal corporate website accessible to all branches.
- **Internet Simulation:** Simulate real-world internet access using NAT/PAT through a simulated ISP.
- **Security:** Apply Access Control Lists, Port Security, and SSH-based device hardening.

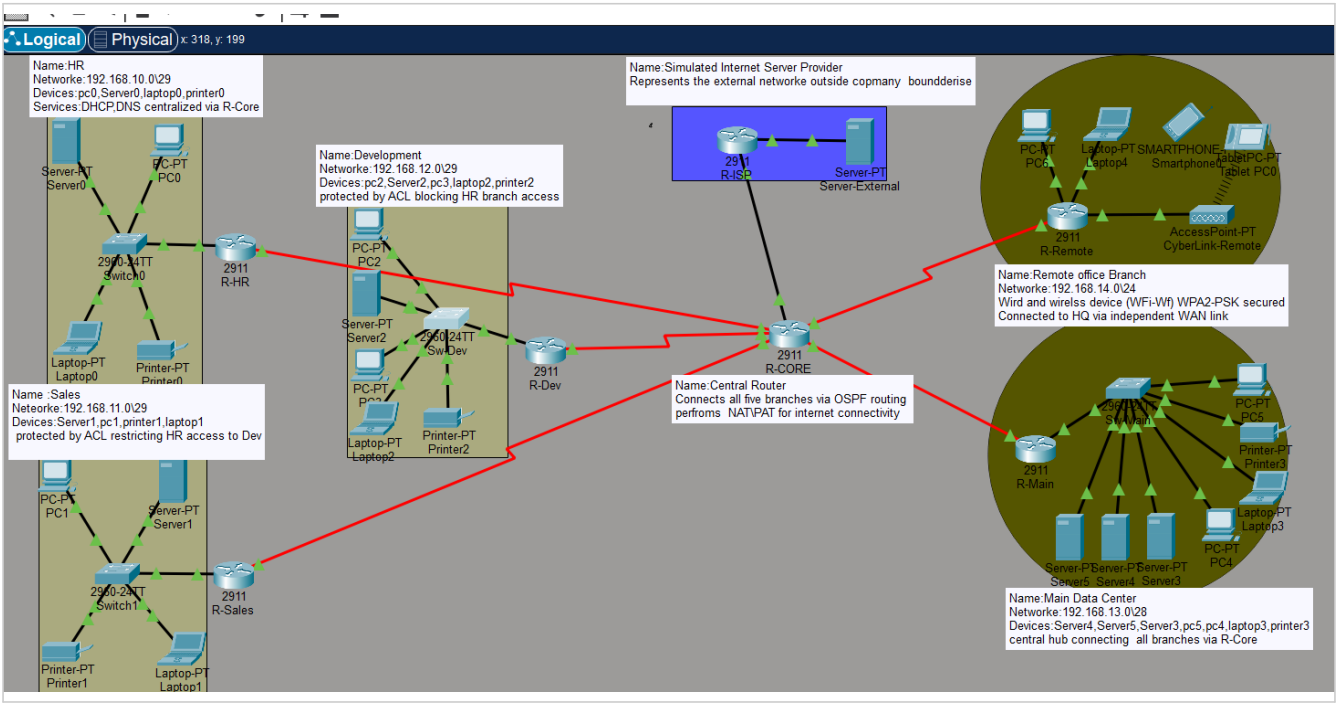


Figure 1 — Full project logical topology

2 Network Topology Overview

The network follows a **Hub-and-Spoke** topology. A central router, **R-CORE**, connects to each branch router over a dedicated point-to-point WAN link. R-CORE also connects outward to **R-ISP**, which simulates the public internet, through a link representing the service provider connection.

Core Infrastructure Components

Device	Role
R-CORE	Central router — OSPF hub, NAT gateway
R-HR / R-Sales / R-Dev	Branch routers for HR, Sales, and Development
R-Main	Main Data Center router — hosts centralized services
R-Remote	Remote Office router — wired & wireless access
R-ISP	Simulated Internet Service Provider edge router

3 Branch Overview

Branch	Network	Devices	Notes
Human Resources (HR)	192.168.10.0/29	PC, Laptop, Server, Printer	Restricted from accessing Development
Sales	192.168.11.0/29	PC, Laptop, Server, Printer	—
Development	192.168.12.0/29	PC, Laptop, Server, Printer	Protected by ACL from HR
Main Data Center	192.168.13.0/28	3 Servers, PCs, Laptop, Printer	Hosts DHCP, DNS, Web services
Remote Office	192.168.14.0/24	PC, Laptop, Tablet, Smartphone	Wi-Fi secured with WPA2-PSK

4 IP Addressing Scheme

The network uses VLSM subnetting to efficiently allocate address space per branch, with /30 point-to-point links for all WAN connections.

LAN Networks

Branch	Network	Subnet Mask	Gateway
HR	192.168.10.0	255.255.255.248	192.168.10.1
Sales	192.168.11.0	255.255.255.248	192.168.11.1
Development	192.168.12.0	255.255.255.248	192.168.12.1
Main	192.168.13.0	255.255.255.240	192.168.13.1
Remote Office	192.168.14.0	varies (/30, /29)	192.168.14.1 / .5 / .9

WAN Links (Point-to-Point /30)

Link	Network
R-HR ↔ R-CORE	10.0.0.0/30
R-Sales ↔ R-CORE	10.0.0.4/30
R-Dev ↔ R-CORE	10.0.0.8/30
R-Main ↔ R-CORE	10.0.0.12/30
R-Remote ↔ R-CORE	10.0.0.16/30
R-CORE ↔ R-ISP	200.100.100.0/30

5 Dynamic Routing (OSPF)

All routers run **OSPF Area 0** (single-area), allowing every router to learn all branch networks dynamically. Each router advertises its directly connected networks, and R-CORE originates a default route toward R-ISP for internet-bound traffic, which the branch routers learn through OSPF.

`router ospf 1`	`network ... area 0`	`default-information originate`

```
R-CORE#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter are
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 0.0.0.0 to network 0.0.0.0

     10.0.0.0/8 is variably subnetted, 10 subnets, 2 masks
C       10.0.0.0/30 is directly connected, Serial0/3/1
L       10.0.0.2/32 is directly connected, Serial0/3/1
C       10.0.0.4/30 is directly connected, Serial0/3/0
L       10.0.0.6/32 is directly connected, Serial0/3/0
C       10.0.0.8/30 is directly connected, Serial0/1/0
L       10.0.0.10/32 is directly connected, Serial0/1/0
C       10.0.0.12/30 is directly connected, Serial0/2/0
L       10.0.0.14/32 is directly connected, Serial0/2/0
C       10.0.0.16/30 is directly connected, Serial0/2/1
L       10.0.0.18/32 is directly connected, Serial0/2/1
     192.168.10.0/29 is subnetted, 1 subnets
O       192.168.10.0/29 [110/65] via 10.0.0.1, 01:06:34, Serial0/3/1
     192.168.11.0/29 is subnetted, 1 subnets
O       192.168.11.0/29 [110/65] via 10.0.0.5, 01:06:34, Serial0/3/0
     192.168.12.0/29 is subnetted, 1 subnets
O       192.168.12.0/29 [110/65] via 10.0.0.9, 01:06:34, Serial0/1/0
     192.168.13.0/28 is subnetted, 1 subnets
O       192.168.13.0/28 [110/65] via 10.0.0.13, 01:06:34, Serial0/2/0
     192.168.14.0/24 is variably subnetted, 3 subnets, 2 masks
O       192.168.14.0/30 [110/65] via 10.0.0.17, 01:06:34, Serial0/2/1
O       192.168.14.4/30 [110/65] via 10.0.0.17, 01:06:34, Serial0/2/1
O       192.168.14.8/29 [110/65] via 10.0.0.17, 01:06:34, Serial0/2/1
     200.100.100.0/24 is variably subnetted, 2 subnets, 2 masks
C       200.100.100.0/30 is directly connected, GigabitEthernet0/0
L       200.100.100.1/32 is directly connected, GigabitEthernet0/0
S*   0.0.0.0/0 [1/0] via 200.100.100.2
                is directly connected, GigabitEthernet0/0
```

Figure 2 — "show ip route" output on R-CORE

```
Password:
R-Sales#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 10.0.0.6 to network 0.0.0.0

     10.0.0.0/8 is variably subnetted, 6 subnets, 2 masks
O       10.0.0.0/30 [110/128] via 10.0.0.6, 00:30:49, Serial0/3/0
C       10.0.0.4/30 is directly connected, Serial0/3/0
L       10.0.0.5/32 is directly connected, Serial0/3/0
O       10.0.0.8/30 [110/128] via 10.0.0.6, 00:30:49, Serial0/3/0
O       10.0.0.12/30 [110/128] via 10.0.0.6, 00:30:49, Serial0/3/0
O       10.0.0.16/30 [110/128] via 10.0.0.6, 00:30:49, Serial0/3/0
     192.168.10.0/29 is subnetted, 1 subnets
O       192.168.10.0/29 [110/129] via 10.0.0.6, 00:30:39, Serial0/3/0
     192.168.11.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.11.0/29 is directly connected, GigabitEthernet0/0
L       192.168.11.1/32 is directly connected, GigabitEthernet0/0
     192.168.12.0/29 is subnetted, 1 subnets
O       192.168.12.0/29 [110/129] via 10.0.0.6, 00:30:39, Serial0/3/0
     192.168.13.0/28 is subnetted, 1 subnets
O       192.168.13.0/28 [110/129] via 10.0.0.6, 00:30:39, Serial0/3/0
     192.168.14.0/24 is variably subnetted, 3 subnets, 2 masks
O       192.168.14.0/30 [110/129] via 10.0.0.6, 00:30:39, Serial0/3/0
O       192.168.14.4/30 [110/129] via 10.0.0.6, 00:30:39, Serial0/3/0
O       192.168.14.8/29 [110/129] via 10.0.0.6, 00:30:39, Serial0/3/0
O*E2 0.0.0.0/0 [110/1] via 10.0.0.6, 00:30:49, Serial0/3/0

R-Sales#
```

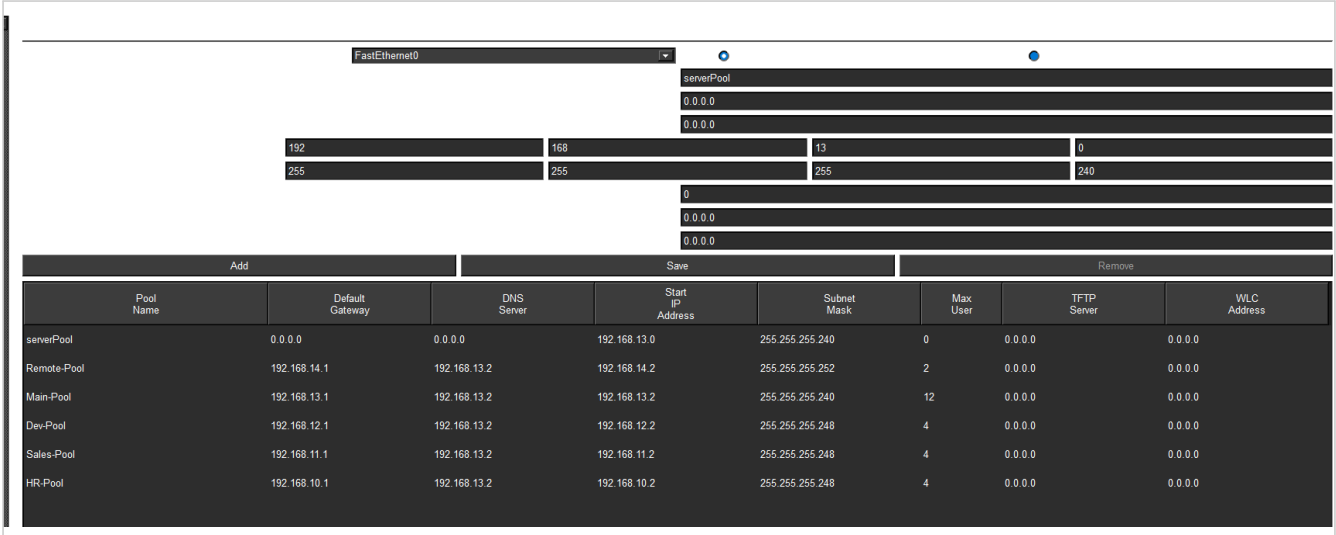
Figure 3 — "show ip route" on R-Sales: all branch networks and the default route learned via OSPF

6 Centralized DHCP Services

A DHCP server is deployed on **Server5** within the Main Data Center, providing automatic IP assignment for all five branches. Since the server resides remotely from each branch's LAN, the **ip helper-address** command relays DHCP broadcasts across the routed network to the central server.

DHCP Pools Configured

Pool	Gateway	DNS Server	Range Start
HR-Pool	192.168.10.1	192.168.13.2	192.168.10.2
Sales-Pool	192.168.11.1	192.168.13.2	192.168.11.2
Dev-Pool	192.168.12.1	192.168.13.2	192.168.12.2
Main-Pool	192.168.13.1	192.168.13.2	192.168.13.2
Remote-Pool	192.168.14.1	192.168.13.2	192.168.14.2



Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
serverPool	0.0.0.0	0.0.0.0	192.168.13.0	255.255.255.240	0	0.0.0.0	0.0.0.0
Remote-Pool	192.168.14.1	192.168.13.2	192.168.14.2	255.255.255.252	2	0.0.0.0	0.0.0.0
Main-Pool	192.168.13.1	192.168.13.2	192.168.13.2	255.255.255.240	12	0.0.0.0	0.0.0.0
Dev-Pool	192.168.12.1	192.168.13.2	192.168.12.2	255.255.255.248	4	0.0.0.0	0.0.0.0
Sales-Pool	192.168.11.1	192.168.13.2	192.168.11.2	255.255.255.248	4	0.0.0.0	0.0.0.0
HR-Pool	192.168.10.1	192.168.13.2	192.168.10.2	255.255.255.248	4	0.0.0.0	0.0.0.0

Figure 4 — DHCP pool configuration on Server5

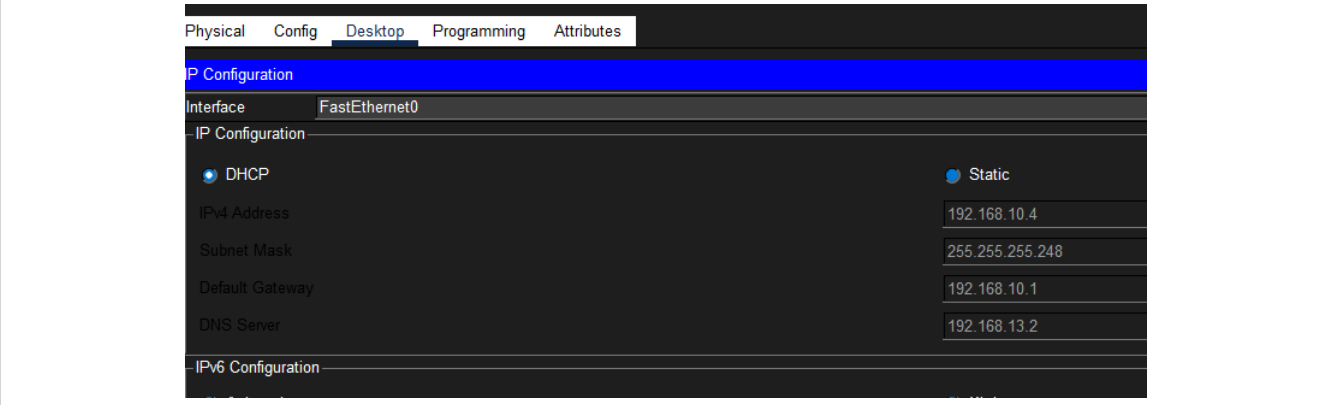


Figure 5 — HR client set to DHCP and receiving 192.168.10.4 with the correct gateway and DNS server

7 DNS Services

A DNS server on **Server3** (192.168.13.2) resolves friendly domain names to IP addresses for both internal and external resources, allowing users to browse by name instead of memorizing raw IP addresses.

Domain Name	Resolves To	Purpose
www.netguard.com	192.168.13.2	Internal corporate website
www.globalnet.com	200.100.200.2	Simulated external internet site

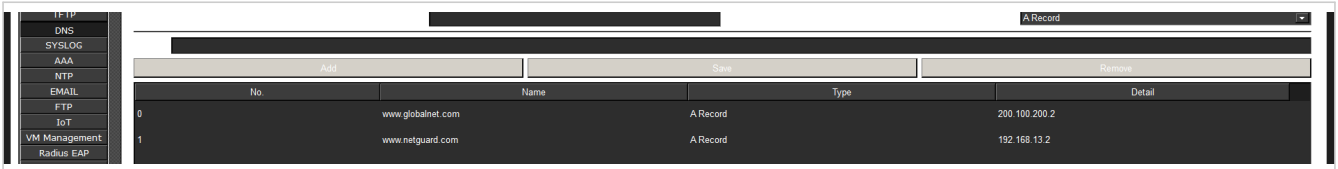


Figure 6 — DNS records configuration on Server3

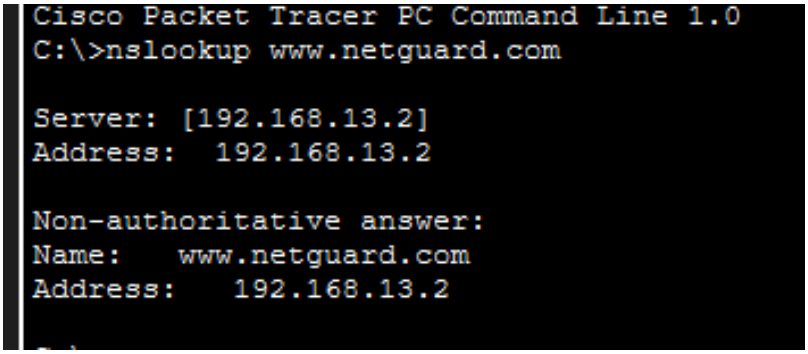


Figure 7 — nslookup resolving www.netguard.com

8 Internal Web Hosting

Server3 in the Main Data Center also hosts the company's internal website, styled with the CyberLink Networks brand identity and accessible from any branch via the internal DNS record.

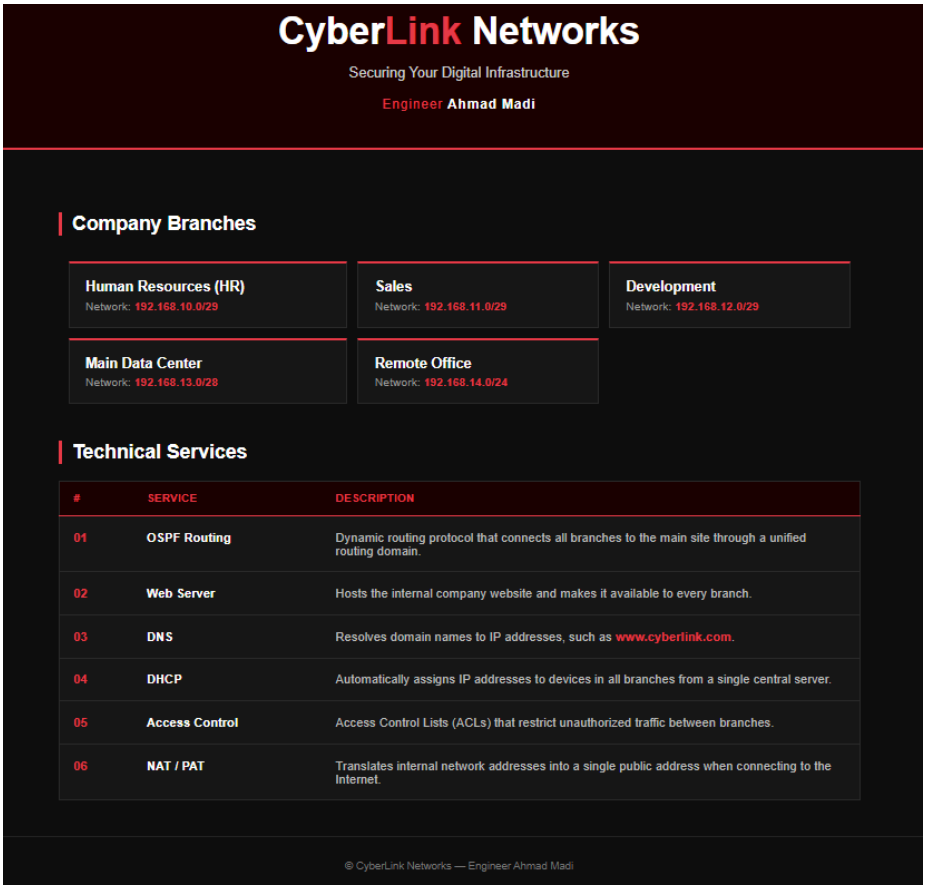


Figure 8 — Internal corporate website opened through www.netguard.com

9 Internet Access Simulation (NAT / PAT)

To simulate real internet connectivity, R-CORE performs **NAT Overload (PAT)**, translating all private internal addresses to its single public-facing IP (**200.100.100.1**) when communicating with the simulated ISP network through R-ISP. An external server behind R-ISP represents a public internet resource.

`ip nat inside`	`ip nat outside`	`ip nat inside source list 1 interface Gi0/0 overload`

```
R-CORE#show ip nat translations
Pro  Inside global     Inside local       Outside local      Outside global
tcp 200.100.100.1:1024 192.168.13.3:1025  200.100.200.2:80   200.100.200.2:80
tcp 200.100.100.1:1025 192.168.13.2:1025  200.100.200.2:80   200.100.200.2:80
tcp 200.100.100.1:1027 192.168.13.2:1027  200.100.200.2:80   200.100.200.2:80
tcp 200.100.100.1:1029 192.168.13.2:1029  200.100.200.2:80   200.100.200.2:80
tcp 200.100.100.1:1030 192.168.13.2:1030  200.100.200.2:80   200.100.200.2:80
tcp 200.100.100.1:1032 192.168.13.2:1032  200.100.200.2:80   200.100.200.2:80
tcp 200.100.100.1:1037 192.168.13.2:1037  200.100.200.2:80   200.100.200.2:80
```

Figure 9 — "show ip nat translations" with active entries

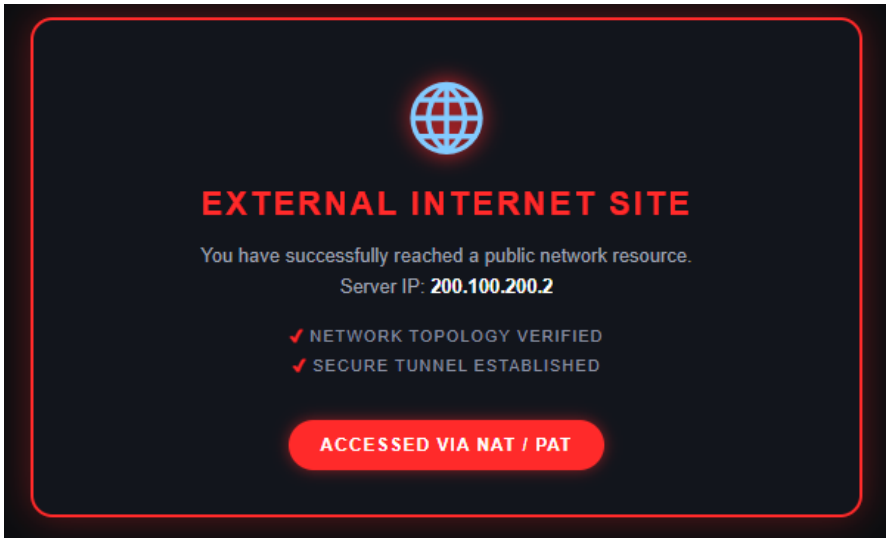


Figure 10 — Client browsing the external simulated website

10 Security Implementation

10.1 Access Control Lists (ACL)

An extended ACL on R-Dev blocks traffic originating from the HR subnet toward the Development subnet, enforcing department-level isolation while allowing all other traffic.

```
Password:
R-Dev#show access-lists
Extended IP access list 101
    10 deny ip 192.168.10.0 0.0.0.7 192.168.12.0 0.0.0.7
    20 permit ip any any (397 match(es))
```

Figure 11 — "show access-lists" on R-Dev

10.2 Port Security

Every access-layer switch port is configured with Port Security, allowing only one learned MAC address per port and shutting the port down automatically if a violation occurs.

```
Switch#show port-security
Secure Port MaxSecureAddr CurrentAddr SecurityViolation Security Action
                (Count)       (Count)         (Count)
------------------------------------------------------------------------
        Fa0/1         1           0                0          Shutdown
        Fa0/2         1           1                0          Shutdown
        Fa0/3         1           0                0          Shutdown
        Fa0/4         1           0                0          Shutdown
        Fa0/5         1           0                0          Shutdown
------------------------------------------------------------------------
```

Figure 12 — "show port-security" output

10.3 Device Hardening (SSH)

Routers are hardened with an encrypted enable secret and SSH-based remote management, replacing insecure Telnet access.

```
R-CORE#show ip ssh
SSH Enabled - version 1.99
Authentication timeout: 120 secs; Authentication retries: 3
```

Figure 13 — "show ip ssh" confirming SSH enabled

10.4 Wireless Security

The Remote Office Access Point uses WPA2-PSK with AES encryption, replacing the default open configuration.

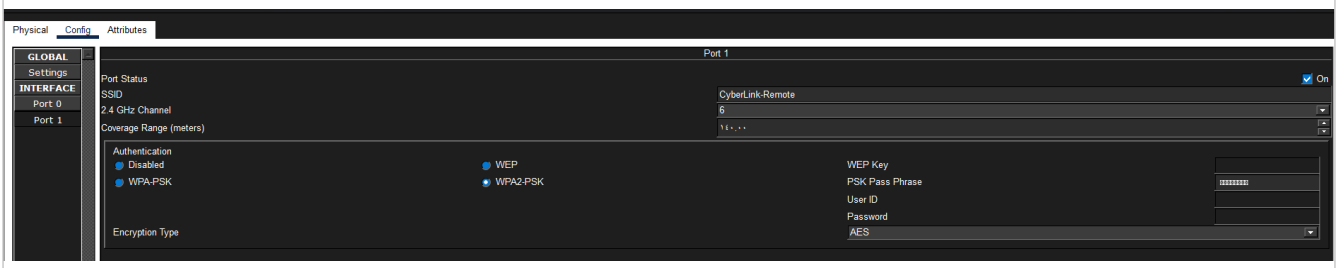


Figure 14 — Access Point configured with WPA2-PSK and AES (passphrase hidden)

11 Testing & Verification Summary

✓	Test	Expected Result
✓	DHCP client request	Automatic IP assignment from correct pool (Figure 5)
✓	DNS resolution	www.netguard.com resolves to 192.168.13.2 (Figure 7)
✓	Internal website access	Website loads via domain name (Figure 8)
✓	External website access (NAT)	Translation entries appear in NAT table (Figures 9, 10)
✓	HR → Development ping	Blocked (Destination host unreachable) (Figure 15)
✓	SSH enabled on routers	SSH enabled on R-CORE (Figure 13)
✓	Port security configured	Max 1 MAC per port, shutdown action (Figure 12)

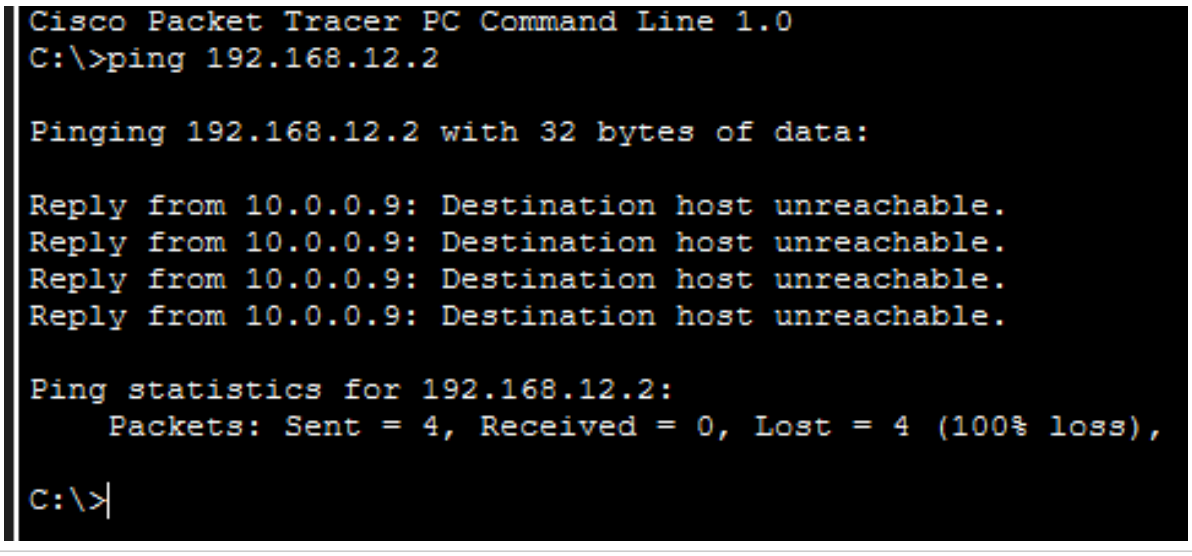


Figure 15 — HR → Development ping blocked by ACL

12 Conclusion

This project successfully delivers a fully functional, secure, multi-branch enterprise network. Through the integration of VLSM addressing, OSPF dynamic routing, centralized DHCP and DNS services, internal web hosting, simulated internet access via NAT/PAT, and a layered security model combining ACLs, Port Security, and SSH hardening, the network reflects practical, industry-aligned enterprise design principles.

The project demonstrates not only core networking competency but also a security-first mindset consistent with the Cyber Security Engineering discipline, making it a strong practical foundation for further study in network defense and infrastructure protection.